Triple Integrals: Solids, Cylindrical Coordinates, and Spherical Coordinates

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§1.1 Triple integrals as weighted volume

A triple integral accumulates a scalar field over a spatial region $\Omega \subseteq \mathbb{R}^3$:

$$\iiint_{\Omega} f(x,y,z)\,dV.$$

If $f = 1$, the integral is volume. If $f = \rho$ is a mass density, the integral is mass. If f is source density, the integral is total source.

Definition 1.1.1. An xy-simple solid has the form

$$\Omega = \{(x,y,z) : (x,y) \in D,\ z_1(x,y) \le z \le z_2(x,y)\}.$$

Then

$$\iiint_{\Omega} f\,dV = \iint_{D} \int_{z_1(x,y)}^{z_2(x,y)} f(x,y,z)\,dz\,dA.$$

There are analogous yz-simple and zx-simple descriptions. If no single description works, split the solid.

§1.2 Rectangular-coordinate example

Example 1.2.1

Let Ω be the tetrahedron

$$x \ge 0, \quad y \ge 0, \quad z \ge 0, \quad x + y + z \le 1.$$

One description is

$$0 \le x \le 1, \qquad 0 \le y \le 1 - x, \qquad 0 \le z \le 1 - x - y.$$

Thus

$$\operatorname{vol}(\Omega) = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\,dz\,dy\,dx = \frac{1}{6}.$$

§1.3 Cylindrical coordinates

Cylindrical coordinates keep the vertical coordinate and use polar coordinates in the horizontal plane:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

The volume element is

$$dV = r\,dr\,d\theta\,dz.$$

They are natural for cylinders, cones around the z-axis, and solids where x^2+y^2 appears.

Example 1.3.1
For the cylinder $x^2 + y^2 \le R^2$, $0 \le z \le h$,

$$V = \int_0^{2\pi}\int_0^R\int_0^h r\,dz\,dr\,d\theta = \pi R^2 h.$$

§1.4 Spherical coordinates

Spherical coordinates are

$$x = \rho\sin\varphi\cos\theta, \qquad y = \rho\sin\varphi\sin\theta, \qquad z = \rho\cos\varphi,$$

where usually $\rho \ge 0$, $0 \le \varphi \le \pi$, and $0 \le \theta < 2\pi$. The volume element is

$$dV = \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta.$$

The factor has a geometric interpretation: small coordinate lengths are approximately

$$d\rho, \qquad \rho\,d\varphi, \qquad \rho\sin\varphi\,d\theta.$$

Multiplying them gives $\rho^2\sin\varphi$.

Example 1.4.1
The ball $x^2 + y^2 + z^2 \le R^2$ has volume

$$\int_0^{2\pi}\int_0^{\pi}\int_0^R \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{4}{3}\pi R^3.$$

§1.5 Integrating a Spherical Sector

A hemisphere and a cone are elementary solids, but their intersection is a useful test of spherical coordinates. Fix $R > 0$ and an angle $0 < \alpha \le \pi$. Let Ω_α be the portion of the ball of radius R lying within polar angle $\varphi \le \alpha$ of the positive z-axis. Its spherical bounds are simply

$$0 \le \rho \le R, \qquad 0 \le \varphi \le \alpha, \qquad 0 \le \theta < 2\pi.$$

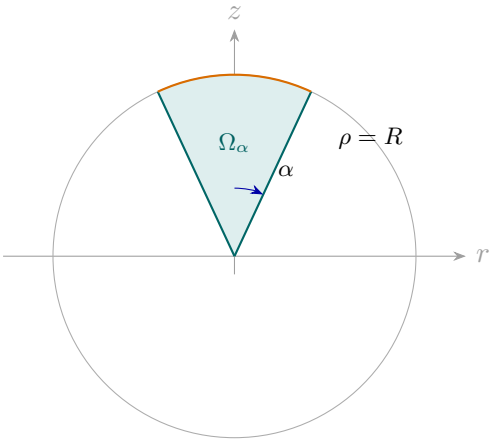


Figure 1.1: A meridional section of a ball cut out by a cone. Rotating the shaded sector about the z-axis produces Ω_α.

For uniform density, its volume is

$$V_\alpha = \int_0^{2\pi}\int_0^{\alpha}\int_0^{R} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$$
$$= \boxed{\frac{2\pi R^3}{3}(1-\cos\alpha)}.$$

At $\alpha = \pi/2$ this is a hemisphere, whose volume is $2\pi R^3/3$; at $\alpha = \pi$ it is the full ball.

Now find its center of mass. Axial symmetry gives $\bar{x} = \bar{y} = 0$. Since $z = \rho\cos\varphi$, the first moment is

$$\iiint_{\Omega_\alpha} z \, dV = \int_0^{2\pi}\int_0^{\alpha}\int_0^{R} \rho^3 \cos\varphi \sin\varphi \, d\rho \, d\varphi \, d\theta$$
$$= \frac{\pi R^4}{4}\sin^2\alpha.$$

Dividing by the volume and simplifying with $\sin^2\alpha = (1-\cos\alpha)(1+\cos\alpha)$, we obtain

$$\boxed{\bar{z} = \frac{3R}{8}(1+\cos\alpha).}$$

For the solid hemisphere $\alpha = \pi/2$, its center of mass lies $3R/8$ above the equatorial plane. A formula involving a variable cone angle has explained a familiar but non-obvious geometric constant.

§1.6 Where the factors r and $\rho^2\sin\varphi$ really come from

Coordinate transformations act on infinitesimal boxes. If T takes parameter coordinates (u, v, w) to Euclidean coordinates, a tiny parameter box is carried approximately to the parallelepiped spanned by the columns of DT. Its volume is multiplied by $|\det DT|$.

For cylindrical coordinates,

$$T(r,\theta,z) = (r\cos\theta, r\sin\theta, z), \qquad |\det DT| = r.$$

For spherical coordinates,

$$T(\rho,\varphi,\theta) = (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi).$$

The coordinate directions are orthogonal, with lengths per unit coordinate increment

$$1, \qquad \rho, \qquad \rho \sin \varphi.$$

Multiplying gives the familiar Jacobian $\rho^2 \sin \varphi$.

A more invariant derivation takes the Euclidean inner products of the coordinate tangent vectors. The resulting metric is

$$ds^2 = d\rho^2 + \rho^2 d\varphi^2 + \rho^2 \sin^2 \varphi \, d\theta^2,$$

so the metric matrix is diagonal with entries $1, \rho^2, \rho^2 \sin^2 \varphi$. Its volume density is

$$\boxed{\sqrt{\det g} = \rho^2 \sin \varphi.}$$

This construction still works for a Riemannian manifold with a curved metric, where there may be no global Cartesian coordinates. The Jacobian formula is the Euclidean instance of integration against the metric volume form.

§1.7 Probability and Spherical Shells

The same Jacobian tells us something unexpected about random vectors. Let

$$X = (X_1, X_2, X_3), \qquad X_i \sim \mathcal{N}(0, 1)$$

independently. Its density in $\mathbb{R}^3$ is

$$f_X(x) = \frac{1}{(2\pi)^{3/2}} e^{-|x|^2/2}.$$

What is the distribution of its distance $R_X = |X|$ from the origin?

The density depends only on radius, but a shell of radius r and thickness dr has volume approximately $4\pi r^2 \, dr$. Integrating over angular coordinates therefore gives

$$\boxed{f_{R_X}(r) = \frac{4\pi r^2}{(2\pi)^{3/2}} e^{-r^2/2}, \qquad r \geq 0.}$$

The factor r^2 is geometry, and the Gaussian factor is probability. Their competition creates a peak away from the origin:

$$\frac{d}{dr} \log f_{R_X}(r) = \frac{2}{r} - r = 0 \quad \Longrightarrow \quad r = \sqrt{2}.$$

Although the three-dimensional Gaussian density f_X is maximal at $x = 0$, its *radius* is most likely near $\sqrt{2}$. There are simply many more points in a larger shell than in a very small one.

In $\mathbb{R}^d$, the radial density of a standard Gaussian is proportional to $r^{d-1} e^{-r^2/2}$, whose mode for $d > 1$ is $\sqrt{d-1}$. Also

$$\mathbb{E}|X|^2 = \sum_{i=1}^{d} \mathbb{E} X_i^2 = d.$$

This is the elementary geometric origin of the concentration of high-dimensional Gaussian mass near a sphere of radius about $\sqrt{d}$. An ordinary triple-integral Jacobian has become the beginning of high-dimensional probability.

§1.8 Divergence measures volume growth of a moving solid

There is another natural question about volume: how does it change as a region moves? Let a vector field F generate a flow Φ_t. The infinitesimal rate of expansion of volume at a point is $\nabla \cdot F$, and in differential-form notation

$$\mathcal{L}_F(dV) = (\nabla \cdot F)\, dV.$$

For the simple radial field $F(x, y, z) = (x, y, z)$, the flow is

$$\Phi_t(p) = e^t p.$$

The linear expansion factor is e^t in each coordinate, so volume is multiplied by e^{3t}. Its instantaneous rate at $t = 0$ is 3, exactly

$$\nabla \cdot F = 1 + 1 + 1 = 3.$$

On a ball of radius R, the divergence theorem gives a numerical identity that can be checked from both sides:

$$\underbrace{\iiint_{|x|\le R} 3\, dV}_{3(4\pi R^3/3)} = \underbrace{\iint_{|x|=R} F \cdot n\, dS}_{R(4\pi R^2)} = 4\pi R^3.$$

On the boundary, $F = Rn$, so the outward flux density is R. The total expansion inside agrees exactly with the amount passing through the boundary.

Finally, volume and orientation must be distinguished. The ordinary change-of-variables formula uses $|\det DT|$ to count positive volume. The oriented form

$$dx \wedge dy \wedge dz$$

pulls back with the signed determinant:

$$T^*(dx \wedge dy \wedge dz) = \det(DT)\, du \wedge dv \wedge dw.$$

For the reflection $T(u, v, w) = (u, -v, w)$, the absolute Jacobian is one but the oriented determinant is -1. Scalar volume remains unchanged, while orientation is reversed. This sign is exactly the data needed when applying Stokes' theorem to oriented boundaries.

The geometric progression is now continuous: describe a solid, choose coordinates adapted to it, derive the Jacobian, integrate mass or volume, and interpret the same factor as the number of possible points in a spherical shell or as the instantaneous change of an oriented volume. The advanced viewpoints do the same bookkeeping as the elementary integral, but in settings where Cartesian coordinates are no longer available.